

Lecture 6

Noise and Hardness: Distributions and the LWE Family

SEC 595: Encrypted Computing

September 27, 2026

Roadmap

Section 1 fixes the four distributions a lattice scheme draws from, and Section 2 states LWE and its ring and module variants, with the short-kernel problem MSIS beside them. How the error grows under addition and multiplication, when decryption is still correct, and where the message sits in a ciphertext are taught inside the scheme lectures that need them: the noise calculus and the correctness template in the BFV lecture, the BGV decryption identity and its failure mode in the BGV lecture, the three-way comparison of message layouts at the end of the CKKS lecture, and the decryption failure probability in the ML-KEM lecture. The security games, the simulation paradigm and Fiat–Shamir are likewise taught where they are first used: IND-CPA and IND-CCA2 in Lecture 9, IND-CPA⁺ in the CKKS lecture, and commitments, Sigma protocols and Fiat–Shamir with aborts in the ML-DSA lecture.

1. The Distributions We Actually Use

Four distributions do all the work. Two are on R_q and are large, two are small: a lattice ciphertext is a large uniform part carrying no information plus a small part carrying everything.

1.1 Uniform on R_q

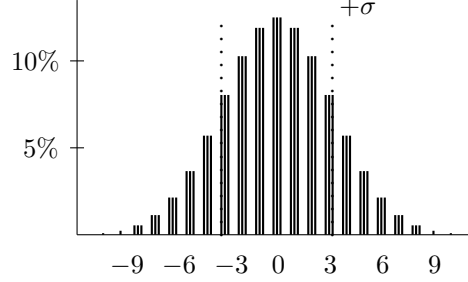
Write $a \leftarrow R_q$ for a ring element whose n coefficients are drawn independently and uniformly from $\mathbb{Z}_q$. If a is uniform and x is any independent ring element then $a + x$ is uniform, since on each coefficient $a \mapsto a + x$ permutes $\mathbb{Z}_q$: uniform plus anything is uniform. ML-KEM and ML-DSA expand a , or the matrix $\mathbf{A}$, from a 32-byte seed with SHAKE-128, so the public key carries the seed.

1.2 The Ternary Key Distribution R_2

The secret key is drawn from R_2 : each coefficient lies in $\{-1, 0, 1\}$, uniformly or with a prescribed Hamming weight h . Then $\|SK\|_\infty = 1$, so SK contributes a factor 1 rather than $\beta \approx 19$ wherever it occurs in a noise bound. In $R = \mathbb{Z}[X]/(X^4 + 1)$ there are $3^4 = 81$ uniform ternary elements, $\binom{4}{2}2^2 = 24$ of weight 2, and $(1, 0, -1, 1) \cdot (1, 1, 0, -1) = (0, 0, 0, -1)$ negacyclically, of norm 1 where the bound $n\|SK\|_\infty\|u\|_\infty = 4$ allows four. A sparse secret with $h \ll n$ cheapens the hybrid dual-lattice attacks of Lecture 5, so deployments using one take $h = 128$ or 192; uniform ternary is the Standard's default.

1.3 The Discrete Gaussian $\mathcal{X}$ and the Bound β

The error distribution is a discrete Gaussian over $\mathbb{Z}$ applied to each coefficient, and β is an integer a sample exceeds only with negligible probability. Every norm argument in the scheme lectures uses β , never the distribution itself.



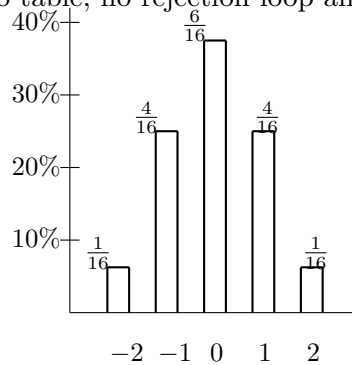
Bars are $\Pr[e = x]$ for $\mathcal{X} = D_{\mathbb{Z},8}$ in percent; the dotted lines mark $\pm\sigma \approx \pm 3.19$, and $\beta = 19$ lies off the picture at six standard deviations.

Definition 1 (Discrete Gaussian and the error bound). For $s > 0$ the discrete Gaussian $D_{\mathbb{Z},s}$ assigns to $x \in \mathbb{Z}$ probability proportional to $\exp(-\pi x^2/s^2)$; its standard deviation is $\sigma = s/\sqrt{2\pi}$. The **error bound** β is an integer such that $e \leftarrow \mathcal{X}$ satisfies $\|e\|_\infty \leq \beta$ with overwhelming probability. The Homomorphic Encryption Standard takes $\mathcal{X} = D_{\mathbb{Z},8}$, $\mu = 0$, $\sigma = 8/\sqrt{2\pi} \approx 3.1915$ and $\beta = \lceil 6\sigma \rceil = \lceil 19.149 \rceil = 19$.

Example 1 (The Standard's numbers, computed). With $s = 8$ the weights $w(x) = \exp(-\pi x^2/64)$ satisfy $\sum_{x \in \mathbb{Z}} w(x) = 8.0000$ to five figures, so $\Pr[e = x] = w(x)/8$, giving 0.12500, 0.11901, 0.10272, 0.08036, 0.05699, 0.03664 and 0.02135 at $x = 0, 1, \dots, 6$. Then $\sigma^2 = \sum_x x^2 \Pr[e = x] = 64/(2\pi) = 10.1859$ and $\sigma = 3.19154 = 8/\sqrt{2\pi} \checkmark$, $\Pr[|e| \leq 9] = 0.99720$, and $\Pr[|e| > 19] = 8.550 \times 10^{-10} = 2^{-30.12}$: one coefficient in 1.2 billion violates $\beta = 19$.

1.4 ML-KEM: the Centered Binomial CBD_η

FIPS 203 draws each coefficient as $\sum_{i=1}^\eta (a_i - b_i)$ from 2η uniform bits: a bit count, a second bit count and a subtraction, with no table, no rejection loop and no data-dependent branch.



Proposition 1. CBD_η is symmetric about 0, supported on $\{-\eta, \dots, \eta\}$, with mean 0 and variance $\eta/2$.

Example 2 (CBD_2 and CBD_3 in full). Each $a_i - b_i$ takes $-1, 0, 1$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$, so summing $\eta = 2$ gives the binomial coefficients of $(1+1)^4$ over 16: $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$, the bars above. Mean $\frac{-2-4+4+2}{16} = 0$, variance $\frac{4}{16} + \frac{4}{16} + 0 + \frac{4}{16} + \frac{4}{16} = 1 = \eta/2 \checkmark$; for $\eta = 3$ the masses are

$\frac{1}{64}, \frac{6}{64}, \frac{15}{64}, \frac{20}{64}, \frac{15}{64}, \frac{6}{64}, \frac{1}{64}$, variance $\frac{3}{2} \checkmark$. ML-KEM-512 uses $\eta_1 = 3, \eta_2 = 2$; ML-KEM-768 and -1024 use $\eta_1 = \eta_2 = 2$.

1.5 Summary of the Four

Uniform on R_q , supported on all of $\mathbb{Z}_q$, gives the public a and $\mathbf{A}$ everywhere. Ternary R_2 , supported on $\{-1, 0, 1\}$, gives SK in BGV, BFV and CKKS, where $\mathcal{X} = D_{\mathbb{Z},8}$ with bound $\beta = 19$ gives e, e_1, e_2 . ML-KEM uses CBD_η with $\eta \in \{2, 3\}$ for both secret and error.

2. The LWE Family

2.1 Search and Decision LWE

An LWE sample is a linear equation in a secret, spoiled by a small error. Search asks for the secret, decision only whether the samples have that form or are uniform; without the error, n samples give the secret by elimination.

Definition 2 (LWE, search and decision). Fix n, q and an error distribution $\mathcal{X}$ on $\mathbb{Z}$, and sample $\mathbf{s} \leftarrow \mathbb{Z}_q^n$. An LWE sample is

$$(\mathbf{a}_i, b_i), \quad \mathbf{a}_i \leftarrow \mathbb{Z}_q^n \text{ uniform}, \quad b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \bmod q, \quad e_i \leftarrow \mathcal{X}.$$

Search LWE is to recover $\mathbf{s}$ from arbitrarily many samples, **decision LWE** to distinguish them from uniform on $\mathbb{Z}_q^n \times \mathbb{Z}_q$. For prime $q = \text{poly}(n)$ the two are equivalent under polynomial-time reductions, and by Lecture 5 an LWE sample is a BDD instance on the q -ary lattice $\Lambda_q(\mathbf{A})$.

Example 3 (Three samples, one secret). Take $q = 13, n = 2, \mathbf{s} = (3, 5)$, errors in $\{-1, 0, 1\}$. Then $\mathbf{a} = (1, 2)$ gives $3 + 10 = 13 \equiv 0$ and, with $e = +1, b = 1$; $(4, 7)$ gives $47 \equiv 8$ and, with $e = -1, b = 7$; $(5, 2)$ gives $25 \equiv 12$ and, with $e = +1, b = 0$. Exhausting all $13^2 = 169$ candidates, the only $\mathbf{s}'$ whose three residuals $b_i - \langle \mathbf{a}_i, \mathbf{s}' \rangle$ all lie in $\{-1, 0, 1\}$ is $(3, 5)$; dropping the third sample leaves nine of them. With the errors set to 0 the first two equations alone give $(3, 5)$, since $\det \begin{pmatrix} 1 & 2 \\ 4 & 7 \end{pmatrix} = -1$ is invertible mod 13.

2.2 Ring-LWE

Ring-LWE replaces the matrix by one element of $R_q = \mathbb{Z}_q[X]/(X^n + 1)$: a sample is $(a, b = a \cdot s + e)$ with a uniform and e drawn coefficientwise from $\mathcal{X}$. One sample is $2n$ elements of $\mathbb{Z}_q$ rather than $n^2 + n$, and the product costs $O(n \log n)$ by the negacyclic NTT of Lecture 3. The lattice is an ideal lattice rather than a random q -ary one.

Example 4 (One Ring-LWE sample by hand). In $R_{13} = \mathbb{Z}_{13}[X]/(X^2 + 1)$, where $X^2 = -1$, take $s = 2 + 3X, a = 5 + X, e = 1 - X$. Then $as = (5 + X)(2 + 3X) = 10 + 17X + 3X^2 = 7 + 17X \equiv 7 + 4X \pmod{13}$, so $b = as + e = 8 + 3X$ and the sample is $(5 + X, 8 + 3X)$: four elements of $\mathbb{Z}_{13}$ carrying a secret of two. Recovering s means finding the s' for which $b - as'$ has both coefficients in $\{-1, 0, 1\}$.

2.3 Module-LWE

Module-LWE takes rank k over R_q instead of rank 1 (Ring-LWE) or rank n over $\mathbb{Z}$ (plain LWE). The lattice has dimension nk , so security is tuned by changing k alone while n, q and every line of NTT arithmetic stay fixed.

Definition 3 (Module-LWE, with Ring-LWE as the case $k = 1$). Fix R_q of degree n and a rank k , and sample $\mathbf{s} \leftarrow \mathcal{X}^k \subseteq R_q^k$ small. Samples are $(\mathbf{A}, \mathbf{b})$ with $\mathbf{A} \leftarrow R_q^{k \times k}$ uniform, $\mathbf{e} \leftarrow \mathcal{X}^k$ and $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e} \in R_q^k$. At $k = 1$ this is Ring-LWE, at $n = 1$ plain LWE; the associated lattice has dimension nk .

Example 5 (ML-KEM's three parameter sets). All three fix $n = 256$, $q = 3329$; only k moves. ML-KEM-512 has $k = 2$, dimension 512, $\eta_1 = 3$, rate 2^{-139} ; ML-KEM-768 has $k = 3$, dimension 768, $\eta_1 = 2$, 2^{-164} ; ML-KEM-1024 has $k = 4$, dimension 1024, $\eta_1 = 2$, 2^{-174} . For ML-KEM-768 the matrix $\mathbf{A}$ is $k^2 = 9$ ring elements, $9 \cdot 256 = 2304$ coefficients of $\mathbb{Z}_{3329}$, all from a 32-byte seed — so the public key sends 32 bytes where the coefficients would need $2304 \cdot 12/8 = 3456$.

2.4 Module-SIS

Module-SIS asks for a short nonzero element of the kernel of a random matrix over R_q : the SIS problem of Lecture 5 lifted to modules. Two distinct valid ML-DSA signatures on one message yield such an element, which is why forgery is hard; MLWE separately hides $\mathbf{s}_1$ inside $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$.

Definition 4 (Module-SIS). Given $\mathbf{A} \leftarrow R_q^{k \times \ell}$ uniform and a bound γ , find a nonzero $\mathbf{z} \in R^\ell$ with $\mathbf{A}\mathbf{z} = \mathbf{0}$ in R_q^k and $\|\mathbf{z}\|_\infty \leq \gamma$. Solutions exist for γ large enough by pigeonhole; the problem is finding one.

Example 6 (A hand-sized MSIS instance). In $R_{13} = \mathbb{Z}_{13}[X]/(X^2 + 1)$ take $k = 1$, $\ell = 2$, $\mathbf{A} = \begin{pmatrix} 1 & 5 + 2X \end{pmatrix}$, so a solution is a nonzero (z_1, z_2) with $z_1 + (5 + 2X)z_2 = 0$. With $z_2 = -2 + X$, $(5 + 2X)(-2 + X) = -10 + X + 2X^2 = -12 + X \equiv 1 + X$, so $z_1 = -1 - X$, $\mathbf{z} = (-1 - X, -2 + X)$, $\|\mathbf{z}\|_\infty = 2$ and $(-1 - X) + (1 + X) = 0$ ✓. Exhaustive search over coefficients in $\{-6, \dots, 6\}$ finds no nonzero solution of norm 1, so the instance is solvable exactly when $\gamma \geq 2$; at $\gamma \geq 13$ it is trivial, since $\mathbf{z} = (13, 0)$ works.

2.5 Which Scheme Uses What

ML-KEM (FIPS 203) rests on MLWE, for IND-CPA of the underlying PKE. ML-DSA (FIPS 204) rests on MLWE for key secrecy and MSIS for unforgeability. BGV, BFV and CKKS rest on RLWE, for IND-CPA and their noise budgets — taken mod t in BGV, under Δ in BFV, as precision in CKKS. Regev and dual-Regev use plain LWE, and so does GSW, as approximate eigenvector encryption.

Looking Ahead: Noise and Correctness

Every scheme built on the problems of Section 2 carries an error term in its ciphertexts, and three facts about that error govern everything that follows. *Addition adds errors*, so sums are essentially free. *Multiplication multiplies them*, so the error grows exponentially in the multiplicative depth of a computation. And *decryption succeeds only while the accumulated error stays below a fixed fraction of q* . Because β bounds every fresh error, all three become arithmetic with the norm inequalities of Lecture 3. Lecture 7 applies this to Regev's scheme; the BFV lecture turns it into a full noise calculus, which BGV, CKKS and ML-KEM then reuse.

Summary

1. Four distributions do the work: uniform on R_q , which absorbs anything added to it; ternary R_2 for keys; $\mathcal{X} = D_{\mathbb{Z},8}$ with $(\mu, \sigma, \beta) = (0, 3.1915, 19)$ for errors; and CBD_η for ML-KEM. It is the *bound* $\beta = \lfloor 6\sigma \rfloor$, not the distribution, that makes every later norm argument finite.
2. $\text{LWE} \rightarrow \text{RLWE} \rightarrow \text{MLWE}$ trades structure for size, and MSIS is the short-kernel problem signatures need.
3. Addition adds errors, multiplication multiplies them, and decryption is correct while the error stays below a fraction of q ; the scheme lectures make this precise.

Exercises

1. **(Centered binomial.)** Write out the mass function of CBD_2 , verify mean 0 and variance 1, and do the same for CBD_3 .
2. **(Module shapes.)** For ML-KEM-768 ($n = 256, q = 3329, k = 3$) give the lattice dimension, count the elements of $\mathbb{Z}_q$ in $\mathbf{A}$, say how the public key avoids sending them, and name which of MLWE, MSIS and RLWE is used. Repeat for ML-DSA-65 ($q = 8380417, (k, \ell) = (6, 5)$).